

Marko Misovski

SOFTWARE ENGINEERING STUDENT

Belgrade, Serbia | +381 61 2855122

markomisa123@gmail.com | github.com/markomisa

EDUCATION

University of Belgrade, School of Electrical Engineering

Software Engineering

GPA: **9.04/10**

Relevant coursework: Operating Systems, Concurrent Programming, Data Structures & Algorithms, Databases.

Mathematical Grammar School

High School Diploma

Belgrade, Serbia

2023–2027 (Expected)

Belgrade, Serbia

2019–2023

EXPERIENCE

Undergraduate Teaching Assistant

University of Belgrade, School of Electrical Engineering

2025–2026 (5th–6th semester)

- Explained SQL queries and database concepts during lab sessions.
- Helped students troubleshoot and debug database assignments.

PROJECTS

KVM Hypervisor

C, Linux, KVM

- Built an x86-64 hypervisor with Linux KVM, configurable guest memory and paging, and one vCPU per VM.
- Added copy-on-write guest file operations and interrupt-driven VM communication synchronized with mutexes and condition variables.

Multithreaded RISC-V Kernel

C++, Assembly

- Implemented a RISC-V kernel with thread creation, preemptive scheduling, context switching, and system calls.
- Added semaphores, memory allocation, and console I/O.

Buddy and Slab Memory Allocator

C, xv6, RISC-V

- Implemented buddy allocation for physical pages and slab allocation for kernel objects in the xv6 RISC-V kernel.
- Integrated both allocators into the kernel.

Rivian AutoVision Hackathon (Pothole Detection)

Python, YOLO, Roboflow

- Contributed to YOLO-based pothole detection through model training, backend development, and database integration.
- Placed second with the team at the Rivian AutoVision Hackathon.

Command Line Interpreter

C++

- Implemented a C++ command-line interpreter with built-in commands such as `echo`, `date`, `time`, `touch`, `wc`, `tr`, `head`, `rm`, and `truncate`.
- Added command pipelines, input/output redirection, and batch execution.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL, Assembly.

Core concepts: Operating Systems, Data Structures & Algorithms, Databases, Multithreading.

Technologies: Linux, Git, KVM, Docker.